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2 **Supplementary Information for**
3 **Bridging Microscopic Dynamics and Rheology in the Yielding of Charged Colloidal**
4 **Suspensions**
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8 **This PDF file includes:**
9 Supplementary text
10 Figs. S1 to S24
11 Table S1
12 Captions for Movies S1 to S3
13 References for SI reference citations

14 **Other supplementary materials for this manuscript include the following:**
15 Movies S1 to S3

Supporting Information Text

Characterization

Colloidal suspensions were prepared by dispersing spherical silica particles surface-functionalized by L-cysteine (SiO_x-Cys) particles with a diameter of $d = 116$ nm and a dispersity of $\Delta r/r = 0.07$ into either deionized water or a 100 mM sodium nitrate (NaNO₃) solution. Samples were studied at a volume fraction of 0.48 (Fig.1A), which allows the development of internal structures associated with complex rheological responses while remaining below the jamming threshold. The chemical structure of the L-cysteine (CYS) ligands grafted onto the SiO_x-Cys surface was verified by NMR (Fig.S1), and the grafting density was determined to be 2.3% by weight from TGA measurements. Zeta potential analysis showed a surface potential of -31.8 mV, and the surface charge density was measured as -0.19 $\mu\text{eq}/\text{g}$. Rheological measurements fitted with the Herschel-Bulkley model yielded a dynamic yield stress of 32.7 Pa, consistent with the static yield stress (24.3 Pa), indicating that the repulsive suspension behaves as a simple yield stress fluid (Fig.S8B).

Structural and dynamic characterizations were performed using small angle X-ray scattering (SAXS) and X-ray photon correlation spectroscopy (XPCS) at beamline 8-ID-I of the Advanced Photon Source (APS). The static structure was examined using coherent X-ray scattering intensity ($I(\vec{q}, \tau)$) profiles (Fig.1B), and equilibrium dynamics were assessed through intensity autocorrelation functions (one-time correlation function ($g_2(\vec{q}, \tau)$)) (Fig.S5E). The SAXS profiles showed a Guinier region at $q > 0.005$ $\text{\AA}^{-1}$ and Bessel-type oscillations, consistent with monodisperse spherical silica particles (SiO_x) particles coated with CYS. In the salted suspension, increased intensity at $q > 0.003$ $\text{\AA}^{-1}$ indicated aggregation and structure formation due to attractive interactions. The $I(\vec{q}, \tau)$ data were analyzed using a two-Yukawa potential model to extract interaction parameters describing the strength and range of interparticle forces (1). According to the reference (1), the equation for the two-Yukawa potential is given below.

$$\frac{V(r)}{k_B T} = \begin{cases} \infty & (0 < r < R) \\ -K_1 \frac{e^{-Z_1(r-R)}}{r} - K_2 \frac{e^{-Z_2(r-R)}}{r} & (r \geq R) \end{cases} \quad [1]$$

From the fitting shown in Fig. 1C, we obtained the following parameters for the aqueous case:

Table S1. Fitting parameters for the two-Yukawa potential in aqueous and salted cases.

Parameter	Aqueous Case	Salted Case
R [nm]	59.1	58.3
K_1	10.81	4.2
K_2	-27.7	-0.12
Z_1	0.13	0.13
Z_2	0.14	0.14

The resulting interaction potentials (inset of Fig.1C, Fig. S3) and structure factors (Fig.1C) show a clear transition from repulsive to attractive interactions with the addition of NaNO₃. Since the potentials in the aqueous and salted suspensions are purely repulsive and purely attractive, respectively, over the entire range of r , there is no energy barrier—that is, no potential peak that must be overcome to reach a more stable state.

The system's dynamics were quantified using the intensity-intensity autocorrelation function (two-time temporal autocorrelation function of intensity ($c_2(\vec{q}, t_1, t_2)$)), derived from 2D raw detector images (Fig.S5C, D):

$$c_2(\vec{q}, t_1, t_2) = \frac{\mathbb{E}[I(\vec{q}, t_1)I(\vec{q}, t_2)]}{\mathbb{E}[I(\vec{q}, t_1)]\mathbb{E}[I(\vec{q}, t_2)]} \quad [2]$$

The resulting $c_2(\vec{q}, t_1, t_2)$ matrices exhibited a Toeplitz-like structure, indicating stationary equilibrium conditions. This confirms that the intrinsic dynamics of SiO_x-Cys particles in both suspensions followed Brownian motion. The $c_2(\vec{q}, t_1, t_2)$ was transformed into the $g_2(\vec{q}, \tau)$ function through diagonal averaging (Fig.S5F):

$$g_2(\vec{q}, \tau) = \frac{\mathbb{E}[I(\vec{q}, t)I(\vec{q}, t + \tau)]}{\mathbb{E}[I(\vec{q}, t)]\mathbb{E}[I(\vec{q}, t + \tau)]} \quad [3]$$

The $g_2(\vec{q}, \tau)$ function was fitted using a stretched exponential function:

$$g_2(\vec{q}, \tau) = 1 + \beta e^{-2\left(\frac{\tau}{\tau_o}\right)^\alpha} \quad [4]$$

Here, τ_o is the relaxation time, α is the stretching exponent representing the distribution of relaxation times, and β is the speckle contrast, which depends on the coherence of the incident beam.

The relaxation time τ_o in the attractive suspension was slightly longer than in the repulsive suspension, indicating the formation of larger aggregates with slower dynamics. This observation aligns with SAXS results, confirming that attractive interactions promote aggregate formation. These findings underscore the impact of interparticle potentials on the structural and dynamic properties of colloidal suspensions.

Simulation Details

Dynamics simulations of dense colloidal suspensions were performed using LAMMPS(2), with 107800 particles in a cubic simulation box. The diameter of the particles followed a normal distribution, $\mathcal{N}(\sigma, (0.07\sigma)^2)$, with a fixed volume fraction of 48 %, consistent with experimental conditions. Here, σ represents the average particle diameter (i.e., 116 nm) and serves as the unit length. The interparticle potential was modeled by the DLVO potential by the combination of "colloid" and "yukawa/colloid" pairwise potential in LAMMPS:

$$\begin{aligned}
 U_A = & -\frac{A_H}{6} \left[\frac{2a_1a_2}{r^2 - (a_1 + a_2)^2} + \frac{2a_1a_2}{r^2 - (a_1 - a_2)^2} + \ln \left(\frac{r^2 - (a_1 + a_2)^2}{r^2 - (a_1 - a_2)^2} \right) \right] \\
 U_R = & \frac{A_H}{37800} \frac{\sigma_s^6}{r} \left[\frac{r^2 - 7r(a_1 + a_2) + 6(a_1^2 + 7a_1a_2 + a_2^2)}{(r - a_1 - a_2)^7} \right. \\
 & + \frac{r^2 + 7r(a_1 + a_2) + 6(a_1^2 + 7a_1a_2 + a_2^2)}{(r + a_1 + a_2)^7} \\
 & - \frac{r^2 + 7r(a_1 - a_2) + 6(a_1^2 - 7a_1a_2 + a_2^2)}{(r + a_1 - a_2)^7} \\
 & \left. - \frac{r^2 - 7r(a_1 - a_2) + 6(a_1^2 - 7a_1a_2 + a_2^2)}{(r - a_1 + a_2)^7} \right]
 \end{aligned} \tag{5}$$

$$U_{el} = \frac{A_{el}}{\kappa} e^{-\kappa(r - (a_1 + a_2))}$$

$$U = U_A + U_R + U_{el}, \quad r < r_c$$

where r is the distance between the center of particles, a_1 and a_2 are the radii of the particles, $\sigma_s = 0.05\sigma$ is the diameter of the solvent, and $r_c = 3\sigma$ is the cutoff distance. To model the interaction in the aqueous suspension, the interaction parameters were set to $A_H = 0.1 k_B T$, $A_{el} = 75 k_B T / \sigma$, $\kappa = 6 \sigma^{-1}$. To model the interaction in salted suspension, the interaction parameters were set to $A_H = 15 k_B T$, $A_{el} = 760 k_B T / \sigma$, $\kappa = 35 \sigma^{-1}$. The simulation potential is compared with the two-Yukawa potential obtained by fitting the experimental structure factor in Fig.S6.

The dynamics of colloidal particles is modeled using the Fast Lubrication Dynamics (FLD) algorithm (3, 4). The motion of colloidal particles is described by the Langevin equation:

$$\mathbf{M} \frac{d\mathbf{U}}{dt} = \mathbf{F}^H + \mathbf{F}^B + \mathbf{F}^P \tag{6}$$

where $\mathbf{M}$ are the masses and moments of inertia of particles, $\mathbf{U}$ are the translational and angular velocities, $\mathbf{F}^H$, $\mathbf{F}^B$, and $\mathbf{F}^P$ are the hydrodynamic forces and torques due to the motion of particles, Brownian forces and torques due to thermal fluctuations, and the conservative forces and torques between particles (eq. (5)). The hydrodynamic force and torques on particles in a linear ambient flow are given by:

$$\mathbf{F}^H = -\mathbf{R}_{FU}(\mathbf{U} - \mathbf{U}^\infty) + \mathbf{R}_{FE}\mathbf{E}^\infty \tag{7}$$

where $\mathbf{U}^\infty$ represents the velocity and the angular velocity of the fluid, $\mathbf{E}^\infty$ is the rate of strain tensor of the fluid, $\mathbf{R}_{FU}$ and $\mathbf{R}_{FE}$ are resistance matrices that determine the hydrodynamic forces and torques on the particles due to their motion relative to the fluid and due to the imposed shear flow, respectively. The Brownian forces and torques on the particles are obtained by the fluctuation-dissipation theorem:

$$\langle \mathbf{F}^B \rangle = 0 \tag{8}$$

$$\langle \mathbf{F}^B \mathbf{F}^B \rangle = 2k_B T \mathbf{R}_{FU} / \Delta t$$

where Δt is the integration timestep. The integration timestep was set to $\Delta t = 0.0005\tau$, where τ is the time unit. In the FLD algorithm, the resistance matrix is decomposed into a far-field isotropic term, $\mathbf{R}_0$, which matches the mean mobility of particles from Stokesian Dynamics, and a near-field term, $\mathbf{R}_\delta$, which describes the pairwise lubrication forces and torques between particles. The pairwise lubrication force originates from the relative motion between two close spheres and can be decomposed into the squeeze, shear, and pump modes. (5) More details on FLD simulations can be found elsewhere (3, 4).

The FLD algorithm is implemented in LAMMPS (6) using a combination of "lubricate/poly" and "brownian/poly" pair styles (with bugs fixed in <https://github.com/lammps/lammps/issues/2933>). In our simulations, the dynamic viscosity of the implicit

fluid was set to $\mu = 1 k_B T \tau / \sigma^3$. To avoid the divergence of the lubrication force as the interparticle gap δ approaches zero, a minimum gap size of $\delta_{\min} = 0.02 \sigma$ was imposed in the calculations whenever $\delta < \delta_{\min}$. The lubrication force between two particles was also neglected if their center-to-center separation exceeded 1.25 times the sum of their radii.

Creep simulations under constant shear stress were performed by barostatting the pressure tensor component $P_{xy} = -\sigma_{xy}$ using the Nosé–Hoover barostat (implemented by *"fix nph/sphere"* in LAMMPS), where σ_{xy} is the applied shear stress. Before each creep simulation, the system underwent a pre-shear procedure, similar to experimental protocols. During the pre-shear, a sinusoidal shear strain was applied to the simulation box for 5000τ with an amplitude of 0.5 and a frequency of $0.002 \tau^{-1}$. After pre-shear, a constant stress σ_{xy} is applied, with $\sigma_{xy} = 0.2$ for the aqueous suspension and $\sigma_{xy} = 1.2$ for the salted suspension.

98 **Supplementary Figures**

99 **NMR Result.**

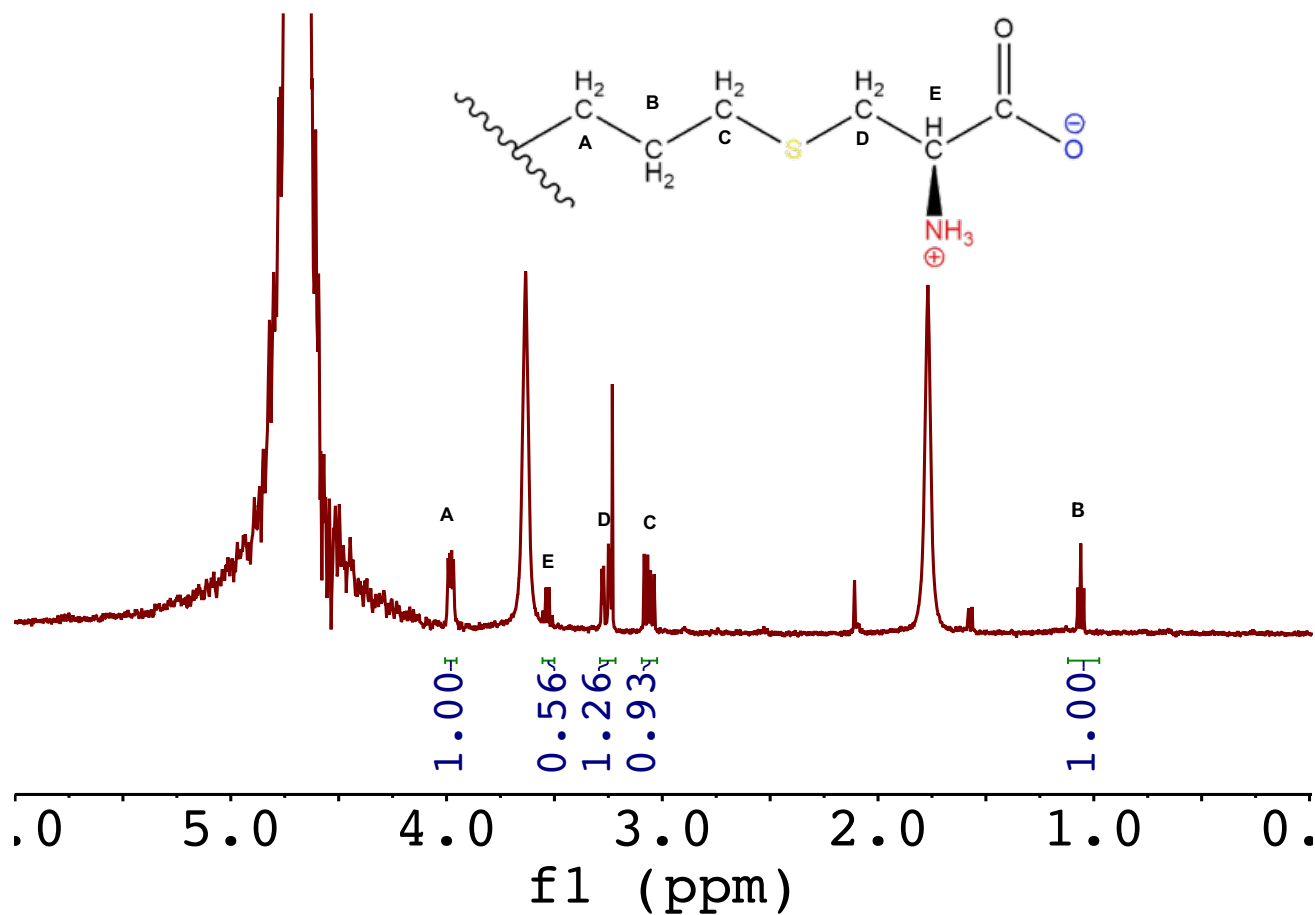


Fig. S1. NMR spectrum of SiO_x-Cys, with characteristic peaks of CYS labeled as D and E.

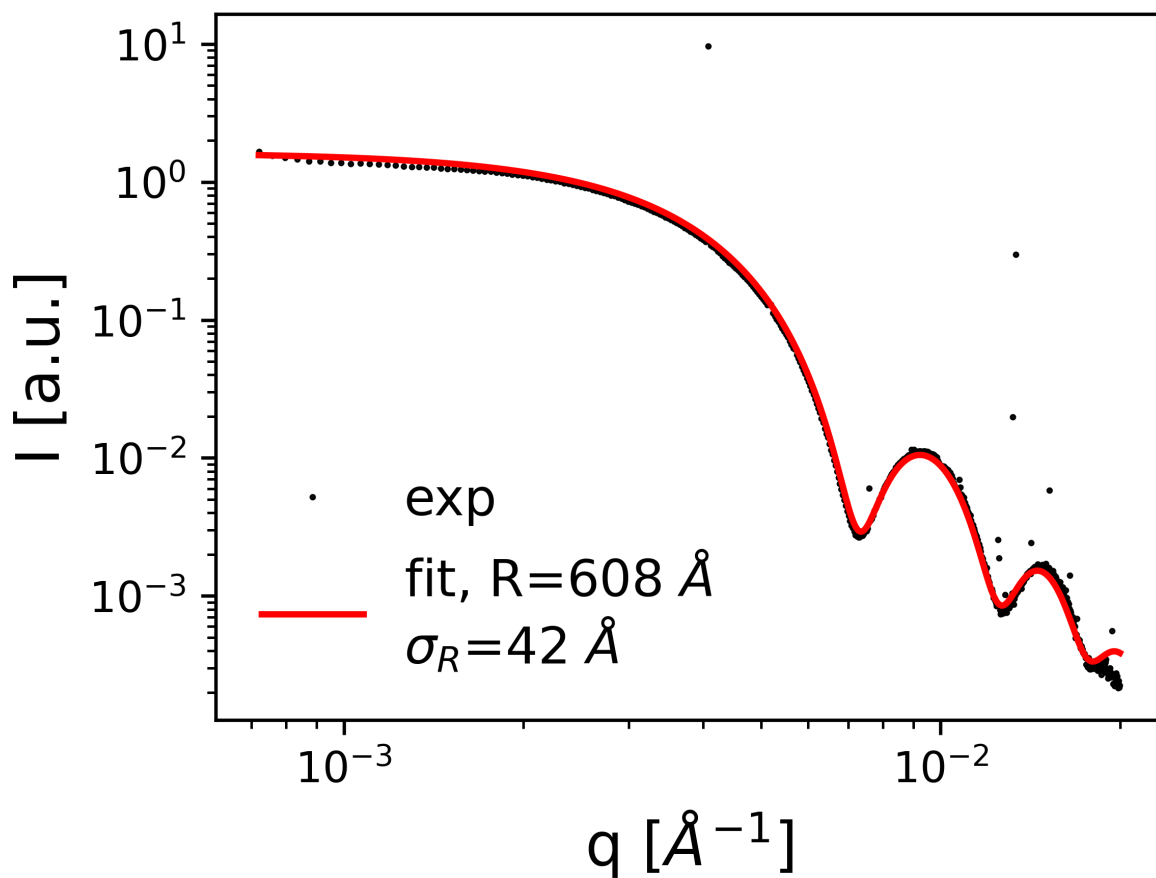


Fig. S2. Fitting of the SAXS scattering profile using the form factor of spherical silica particles. The fitting yields a particle radius of 608 Å (diameter of 122 nm) with a standard deviation σ_R of 42 Å, corresponding to a polydispersity of 0.07. These results indicate that the particle system is nearly monodisperse.

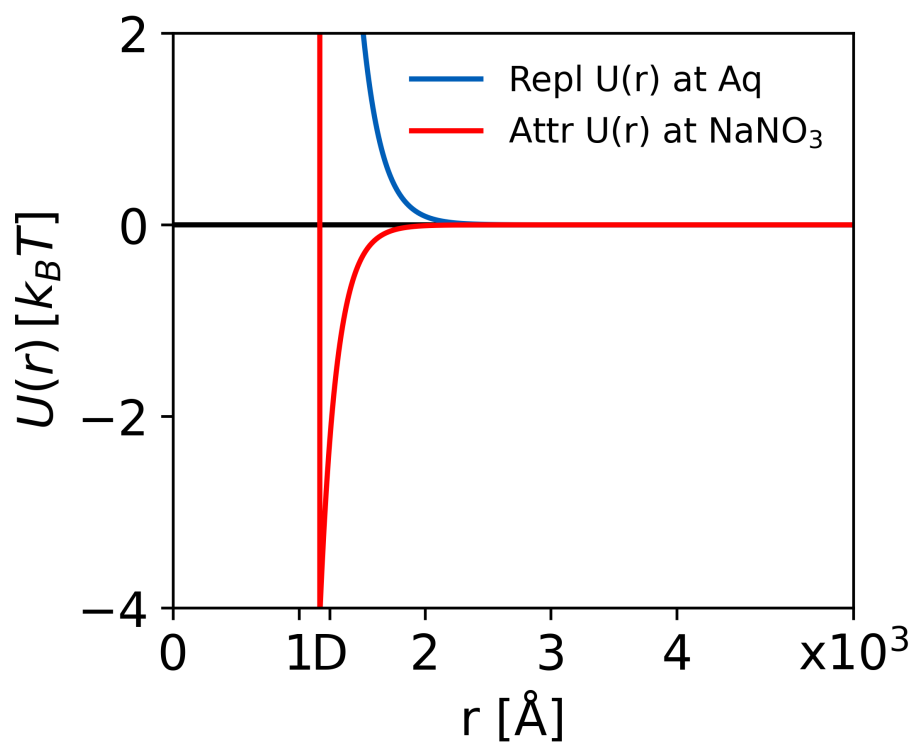


Fig. S3. Zoomed-in view of the surface potential for particles in both aqueous and salted suspensions. In the aqueous suspension, the potential is purely repulsive, while in the salted suspension, it is purely attractive across the entire range of r . In both cases, there is no energy barrier—that is, no potential peak that must be overcome to reach a more stable state.

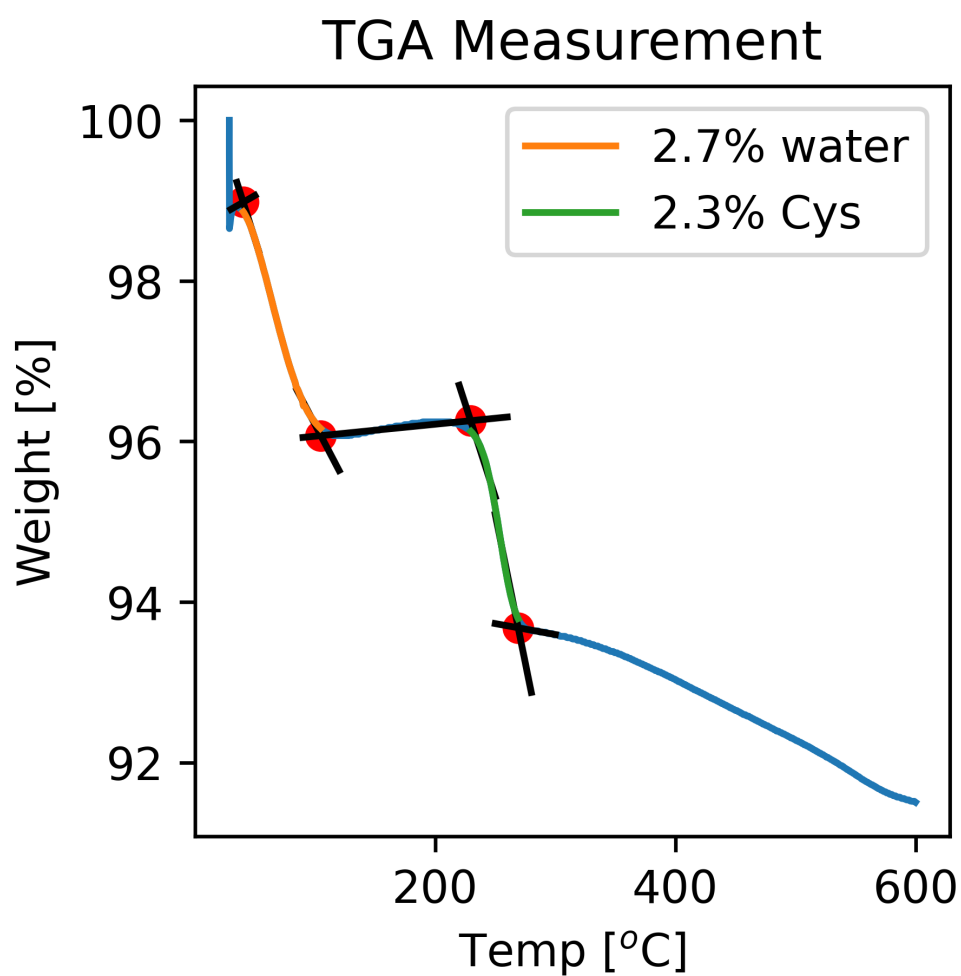


Fig. S4. TGA analysis of $\text{SiO}_x\text{-Cys}$ shows a grafted cysteine content of 2.3% and a water content of 2.7%, confirming the composition and grafting density.

Static Structure and Dynamics. The static structure and equilibrium dynamics measurements demonstrate that the introduction of an attractive potential leads to the formation of internal structures in the system. This is supported by the increased low- q intensity in the SAXS profile, longer particle relaxation times, and a larger hysteresis area observed in shear sweep measurements.

The rheological hysteresis observed during the shear ramp measurement (Fig.S8A) further supports the idea that the addition of salts transforms a simple fluid of repulsive colloids into a system exhibiting complex rheological behavior. The structural and dynamic changes revealed by SAXS and XPCS (Fig.1) correlate with the rheological data, where $\text{SiO}_x\text{-Cys}$ in a salt solution shows significant hysteresis. In contrast, the stress curves during shear sweep up and down are identical in the repulsive suspension, indicating the absence of hysteresis.

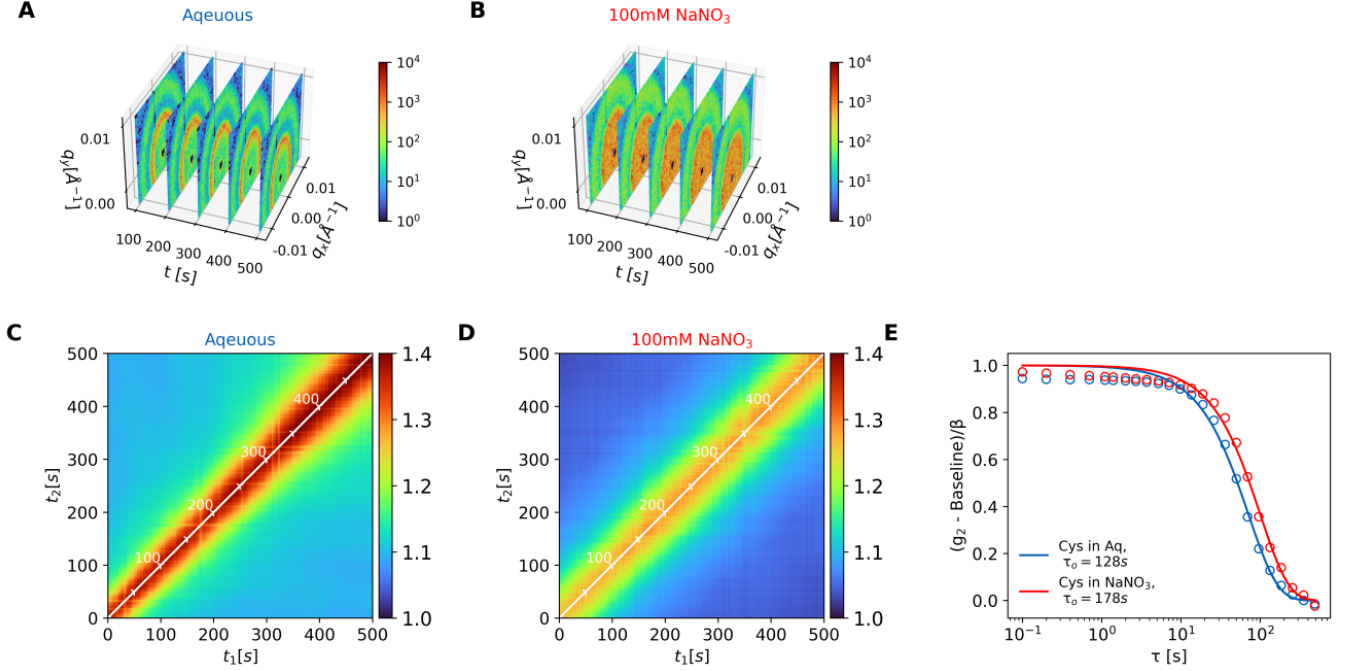


Fig. S5. SAXS and $c_2(\vec{q}, t_1, t_2)$ results for structural and dynamic characterization. (A, B) 2D SAXS patterns are azimuthally integrated to obtain 1D profiles, with experimental data represented as markers in Fig.1. (C, D) Dynamics are analyzed by calculating $c_2(\vec{q}, t_1, t_2)$ (eq. (2)) from 2D SAXS data. The equilibrated state allows for the calculation of $g_2(\vec{q}, \tau)$ through averaging, which is then used to obtain $c_2(\vec{q}, t_1, t_2)$. (E) The $g_2(\vec{q}, \tau)$ function is fitted with a stretched exponential to extract the relaxation time. The introduction of attractive potentials promotes the formation of internal structures, as indicated by the increased low- q intensity in SAXS profiles and longer particle relaxation times.

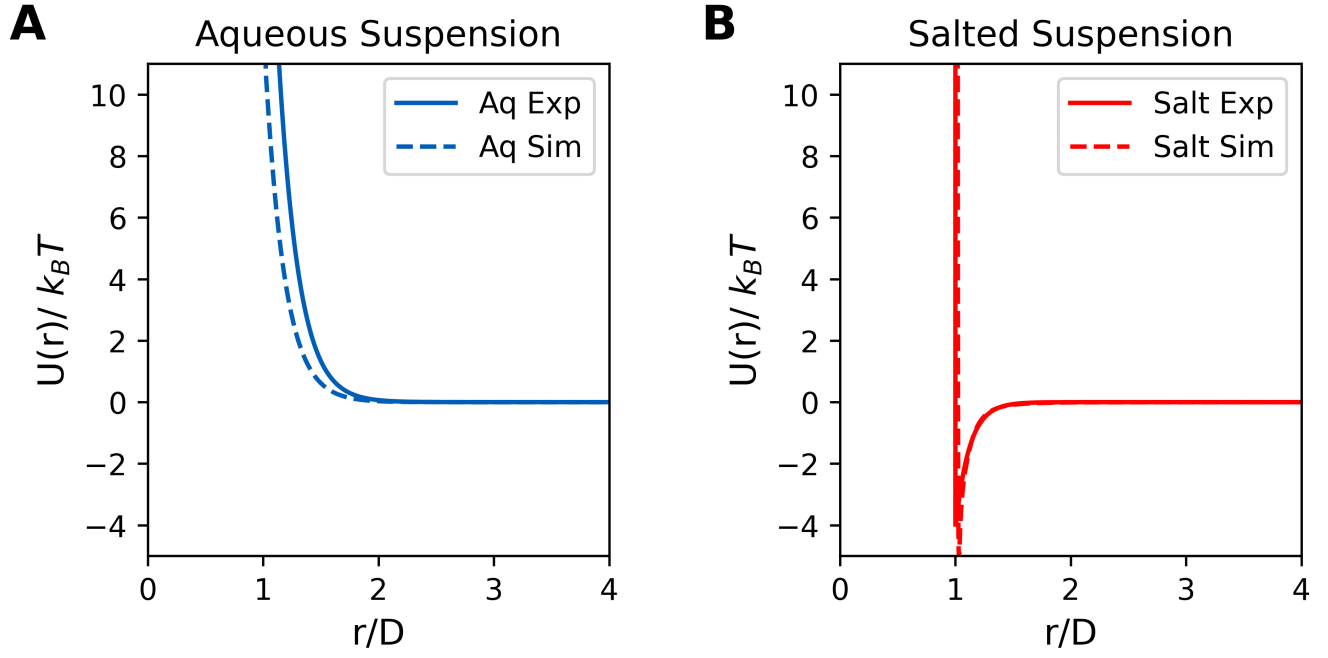


Fig. S6. Simulated SAXS profiles using the two-Yukawa potential for repulsive (**A**) and attractive (**B**) suspensions. The results illustrate the shift in interparticle interactions from purely repulsive to attractive with the addition of salt.

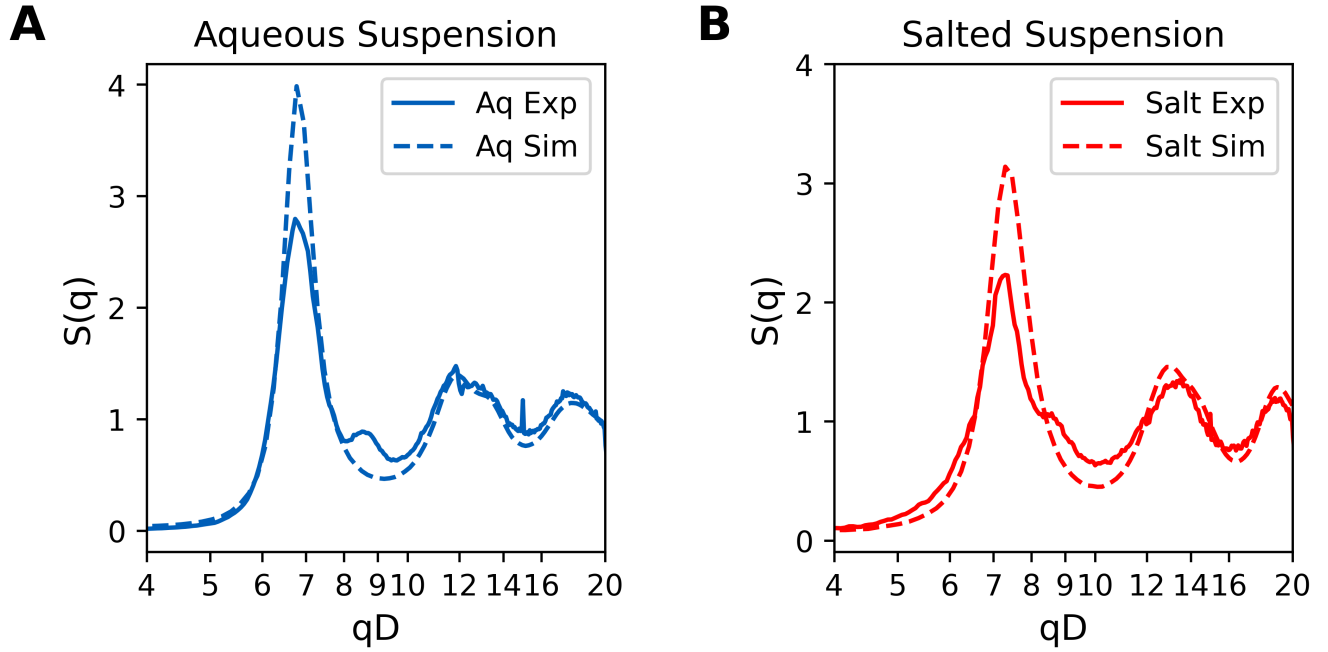


Fig. S7. Comparison of experimental (solid lines) and simulated (dashed lines) structure factors for repulsive (**A**) and attractive (**B**) suspensions. The good agreement confirms that the simulations effectively capture the interparticle interactions and structural features of both systems. The missing peak around $qD = 8.5$ in the simulated results, compared to the experimental data, is attributed to the local presence of minor crystalline phases in the experimental system, whereas the simulated system is completely homogeneous.

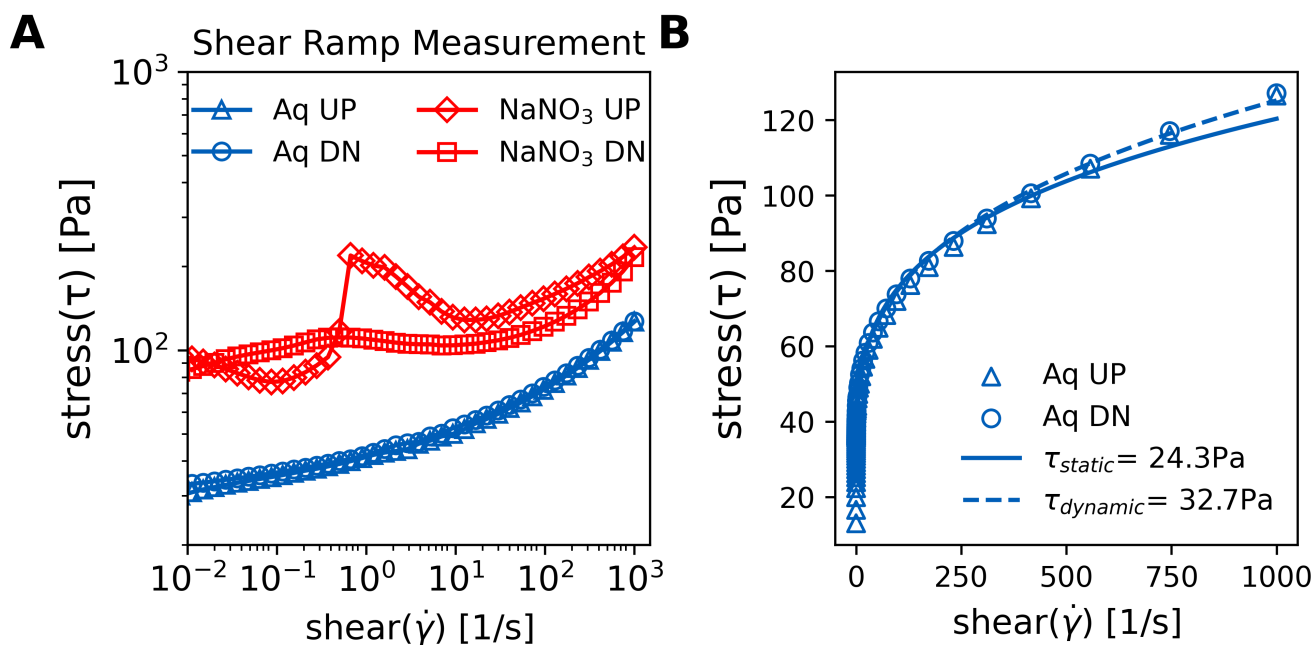


Fig. S8. Shear ramp measurement and modeling with Herschel–Bulkley for suspensions with attractive and repulsive interactions. (A) Shear ramp measurements for an aqueous suspension with a repulsive potential (blue) and a salted suspension with an attractive potential (red). The addition of salt induces rheological hysteresis, evident from the hysteresis area between the ramp-up (UP) and ramp-down (DN) measurements, as the surface potential of particles changes from a repulsive to an attractive interaction potential. **(B)** The stress-shear relationship for the repulsive suspension is fitted using the Herschel–Bulkley model, where the ramp-up measurement gives a static yield stress of 24.3 Pa, and the ramp-down measurement gives a dynamic yield stress of 32.7 Pa. The close consistency between these two values indicates that the repulsive suspension behaves as a simple yielding fluid.

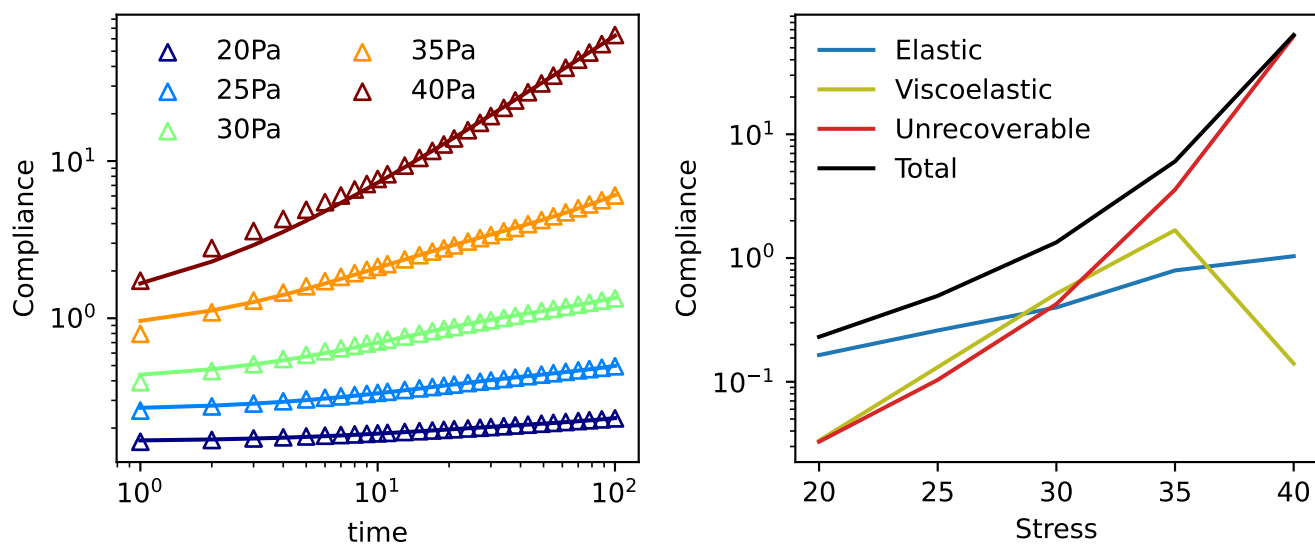


Fig. S9. Creep response of a repulsive suspension fitted using the Burger model. The fitting closely matches the experimental data, indicating that the Burger model effectively captures the creep behavior under applied stress. As stress increases, the response shifts from elastic to viscous dominance.

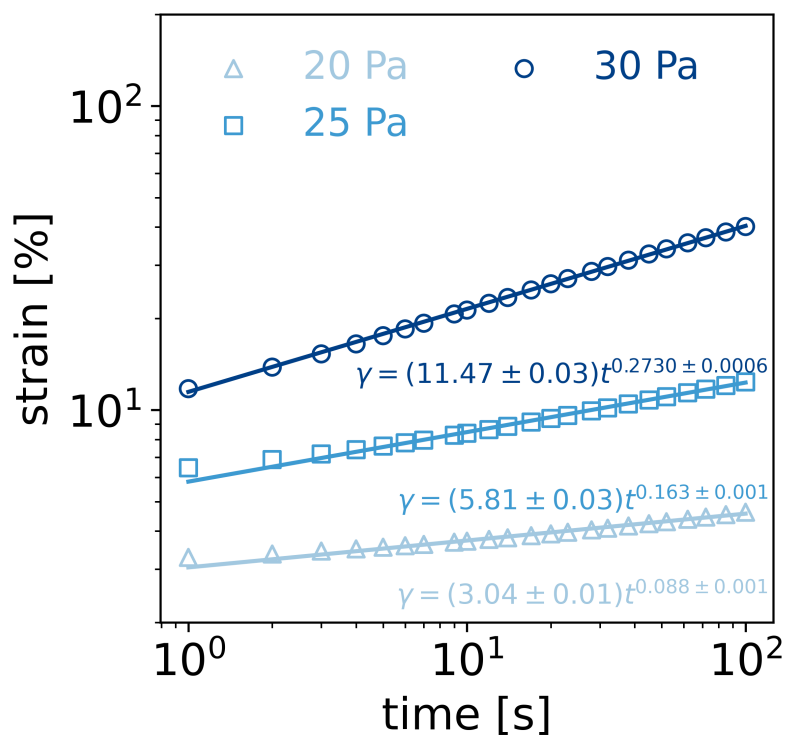


Fig. S10. Fitting results for the strain–time curves of the aqueous suspension under stress of 20, 25, and 30 Pa. Experimental data points are shown as dots, and the solid lines represent the corresponding power-law fits. The curves are fitted using a power-law function characterized by a prefactor (scale) and an exponent. As the applied stress increases, the exponent approaches $1/3$, consistent with the characteristics of Andrade yielding.

Zeta Potential Report
v2.3



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Sample Details

Sample Name: 120nm_SiOx_Cys_01 1
SOP Name: mansettings.nano
General Notes:

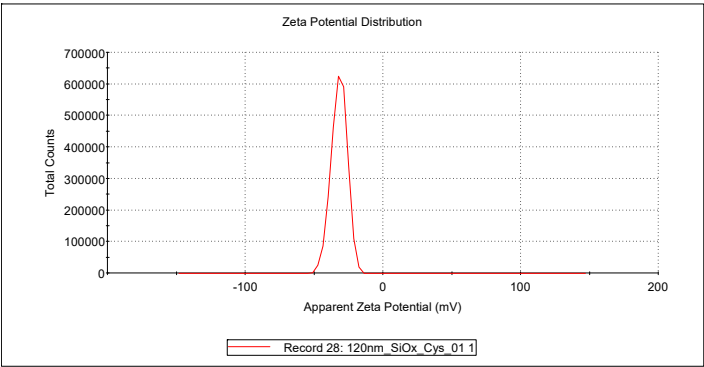
File Name: 120nm_SiOx_Particle.dts Dispersant Name: Water
Record Number: 28 Dispersant RI: 1.330
Date and Time: Tuesday, February 1, 2022 1:58:4... Viscosity (cP): 0.8872
Dispersant Dielectric Constant: 78.5

System

Temperature (°C): 25.0 Zeta Runs: 12
Count Rate (kcps): 334.0 Measurement Position (mm): 2.00
Cell Description: Clear disposable zeta cell Attenuator: 9

Results

	Mean (mV)	Area (%)	St Dev (mV)
Zeta Potential (mV): -31.8	Peak 1: -31.8	100.0	5.71
Zeta Deviation (mV): 5.71	Peak 2: 0.00	0.0	0.00
Conductivity (mS/cm): 0.0158	Peak 3: 0.00	0.0	0.00
Result quality : Good			



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Zetasizer Ver: 7.12
Serial Number: MAL1013322

File name: 120nm_SiOx_Particle.dts
Record Number: 28
01 Feb 2022 2:23:49 PM

Fig. S11. Zeta potential measurements show that the particles are negatively charged, with a monodisperse distribution centered at -31.8 mV

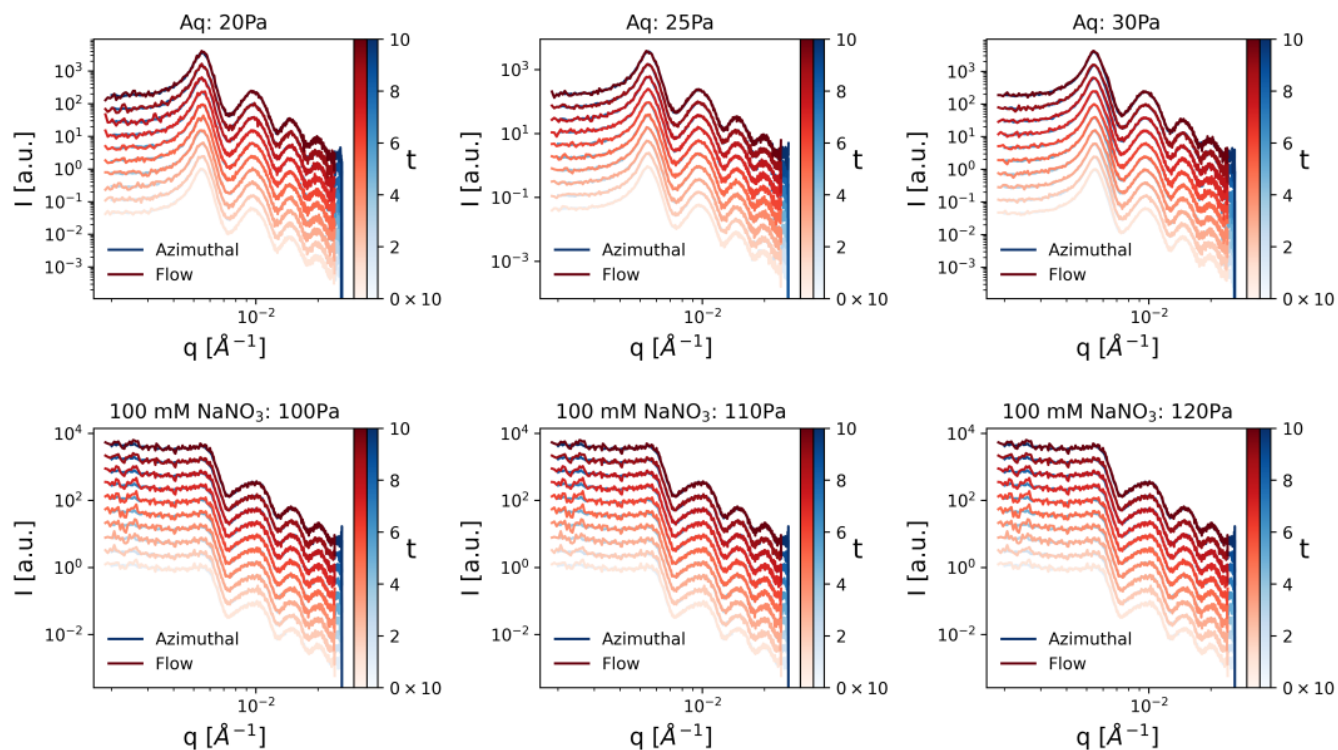


Fig. S12. Flow-direction and azimuthal SAXS profiles were measured during creep tests for both repulsive and attractive suspensions. The agreement between the flow-direction and azimuthal profiles confirms the isotropic and amorphous nature of the system. Furthermore, the lack of significant changes in the SAXS profiles suggests minimal structural evolution under creep conditions.

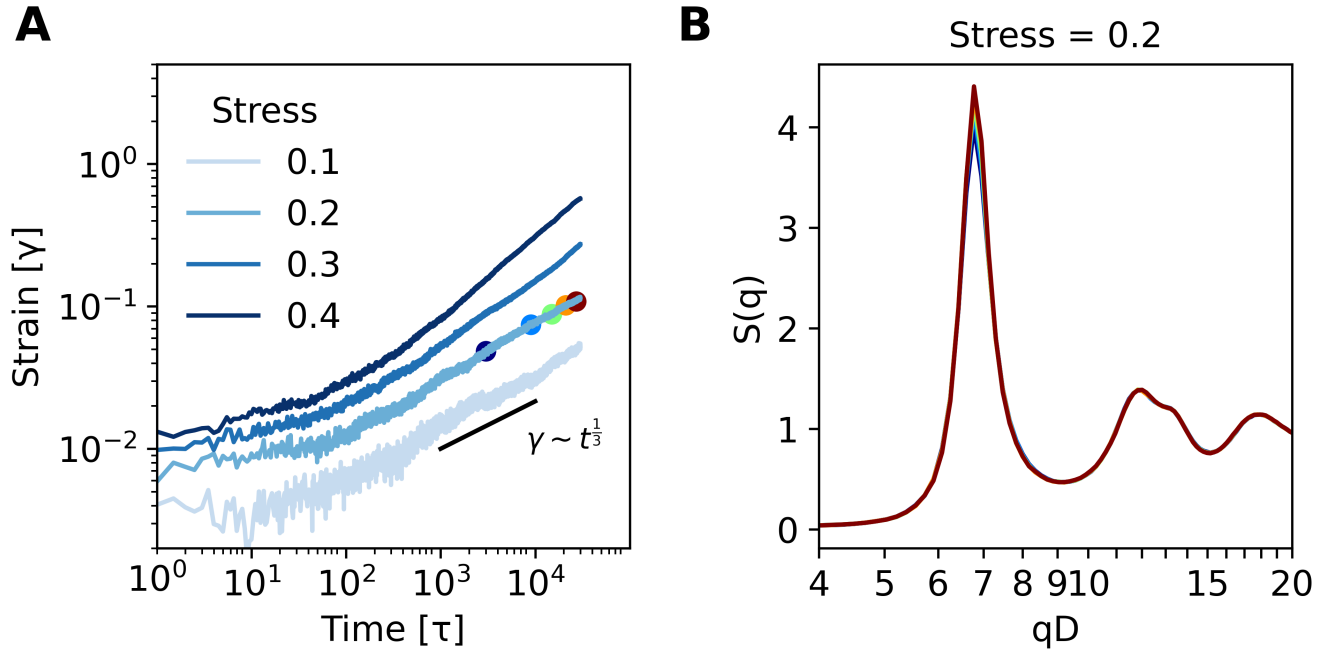


Fig. S13. (A) Creep test results for aqueous suspensions simulated under stress levels ranging from 0.1 to 0.4. The black solid line indicates a strain increase following a power-law relationship, $\gamma \sim t^{\frac{1}{3}}$, corresponding to a shear rate scaling as $\dot{\gamma} \sim t^{-\frac{2}{3}}$. This behavior aligns with the characteristics of Andrade-like creep. (B) Structure factor evolution during the creep test. The test duration is divided into five equal intervals, with structure factors averaged within each interval. The color of each structure factor corresponds to the midpoint of the respective interval, as shown in Panel A. There is a slight increase in the peak intensity of the structure factor during the simulated yielding process. This may be attributed to shear-induced crowding, as suggested in reference (7). However, beyond this minor increase, there is no significant structural change observed.

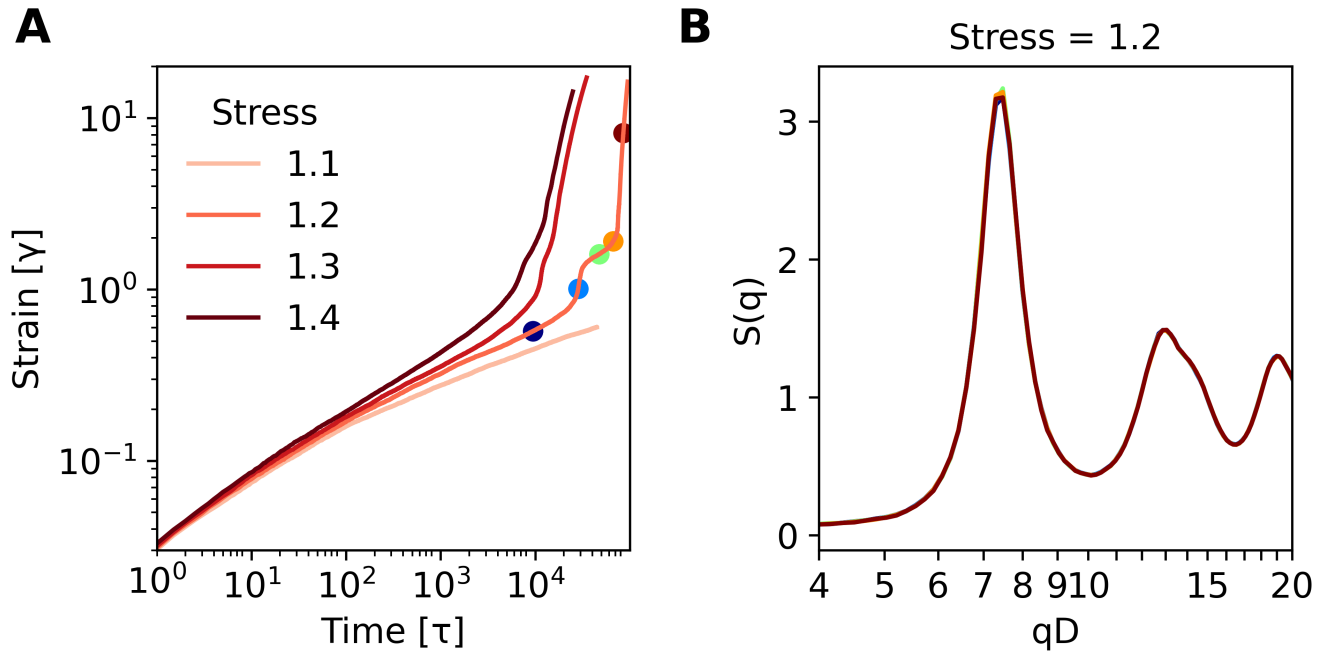


Fig. S14. (A) Simulated creep tests of attractive suspensions under stress levels ranging from 1.1 to 1.4. (B) Structure factors were calculated during the creep tests, with the test duration divided into five equal intervals. Each structure factor is averaged over its respective interval, and the colors correspond to the midpoint of each interval as shown in Panel A. The results indicate that no significant structural changes occurred throughout the creep test.

117 **Analysis of $c_2(\vec{q}, t_1, t_2)$ in all directions.**

118 **Aqueous suspension.** During creep measurements under horizontally applied stress, the $c_2(\vec{q}, t_1, t_2)$ decay exhibits a significantly
 119 faster rate in the horizontal direction. This anisotropic behavior gradually decreases as the observation angle moves away from
 120 the horizontal axis, reaching its slowest decay in the vertical direction (Fig.S15, S16, and S17).

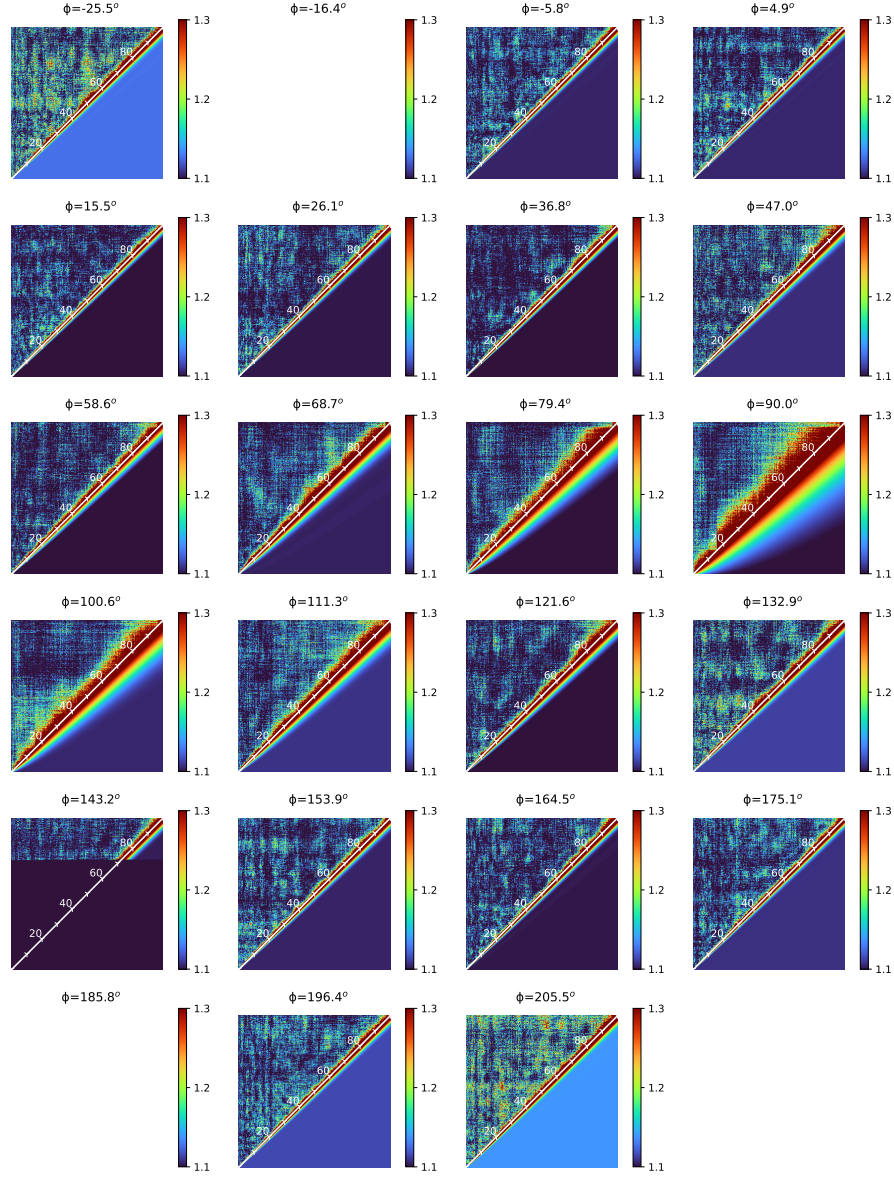


Fig. S15. $c_2(\vec{q}, t_1, t_2)$ functions for SiO_x-Cys in aqueous suspension under a stress of 20 Pa, presented for all directions. Experimental results are shown in the upper corners of each panel, while the corresponding fitting results are displayed in the lower corners.

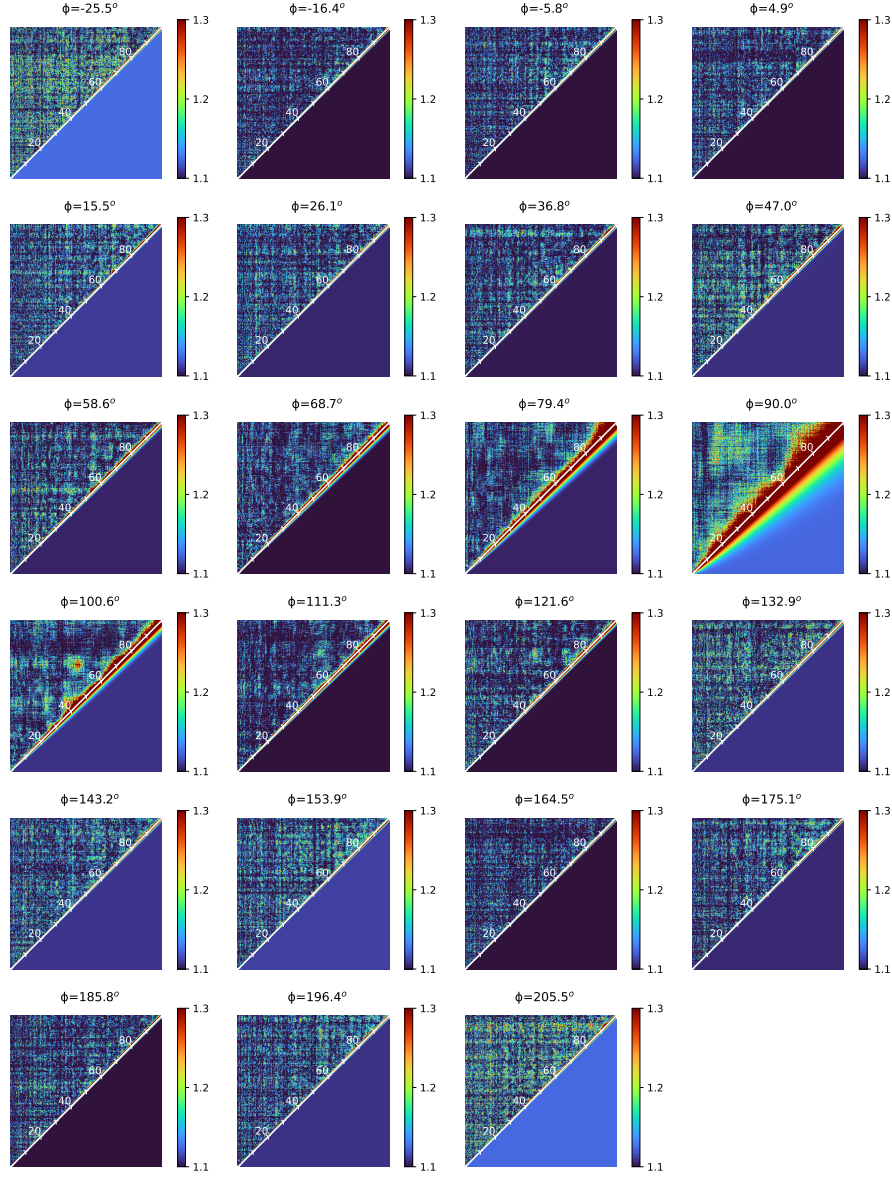


Fig. S16. $c_2(\vec{q}, t_1, t_2)$ functions for $\text{SiO}_x\text{-Cys}$ in aqueous suspension under a stress of 25 Pa, presented for all directions. Experimental results are shown in the upper corners of each panel, while the corresponding fitting results are displayed in the lower corners.

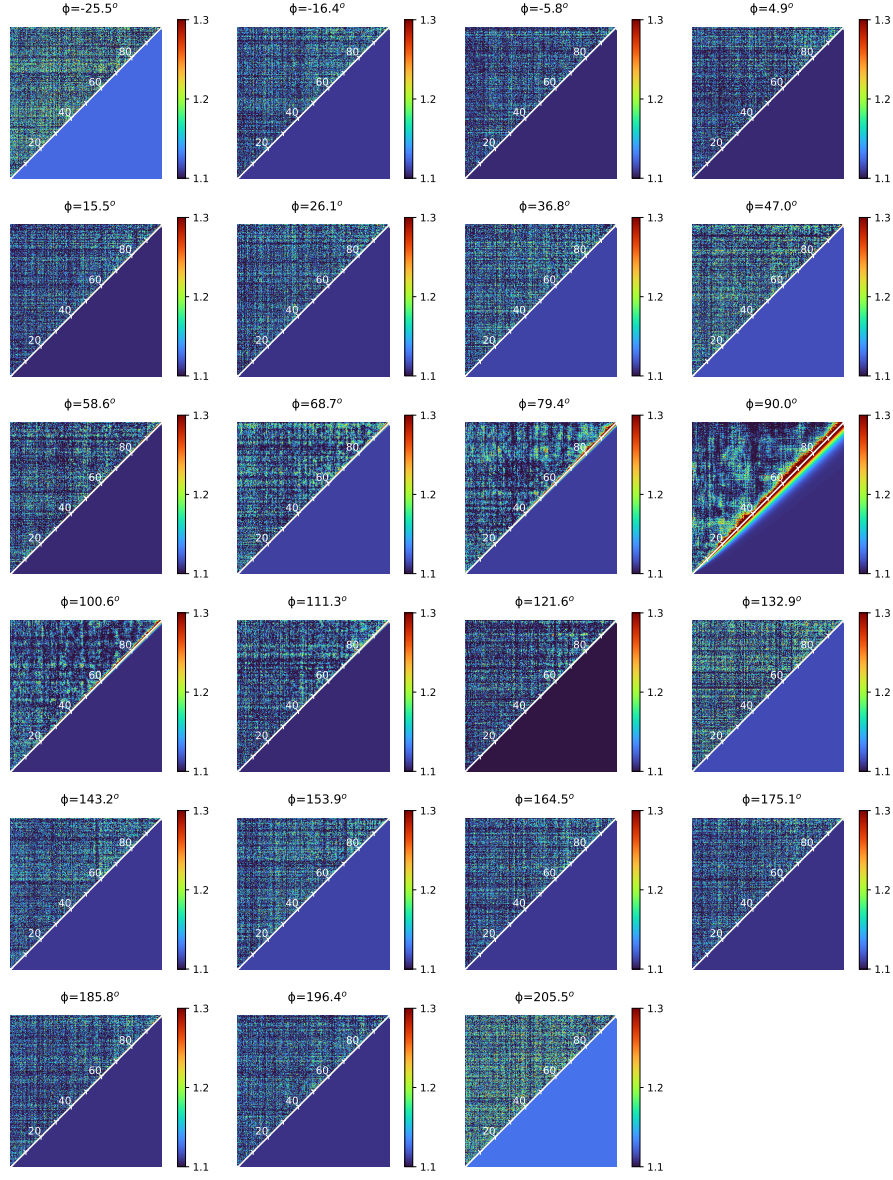


Fig. S17. $c_2(\vec{q}, t_1, t_2)$ functions for $\text{SiO}_x\text{-Cys}$ in aqueous suspension under a stress of 30 Pa, presented for all directions. Experimental results are shown in the upper corners of each panel, while the corresponding fitting results are displayed in the lower corners.

Suspension in 100 mM NaNO_3 .

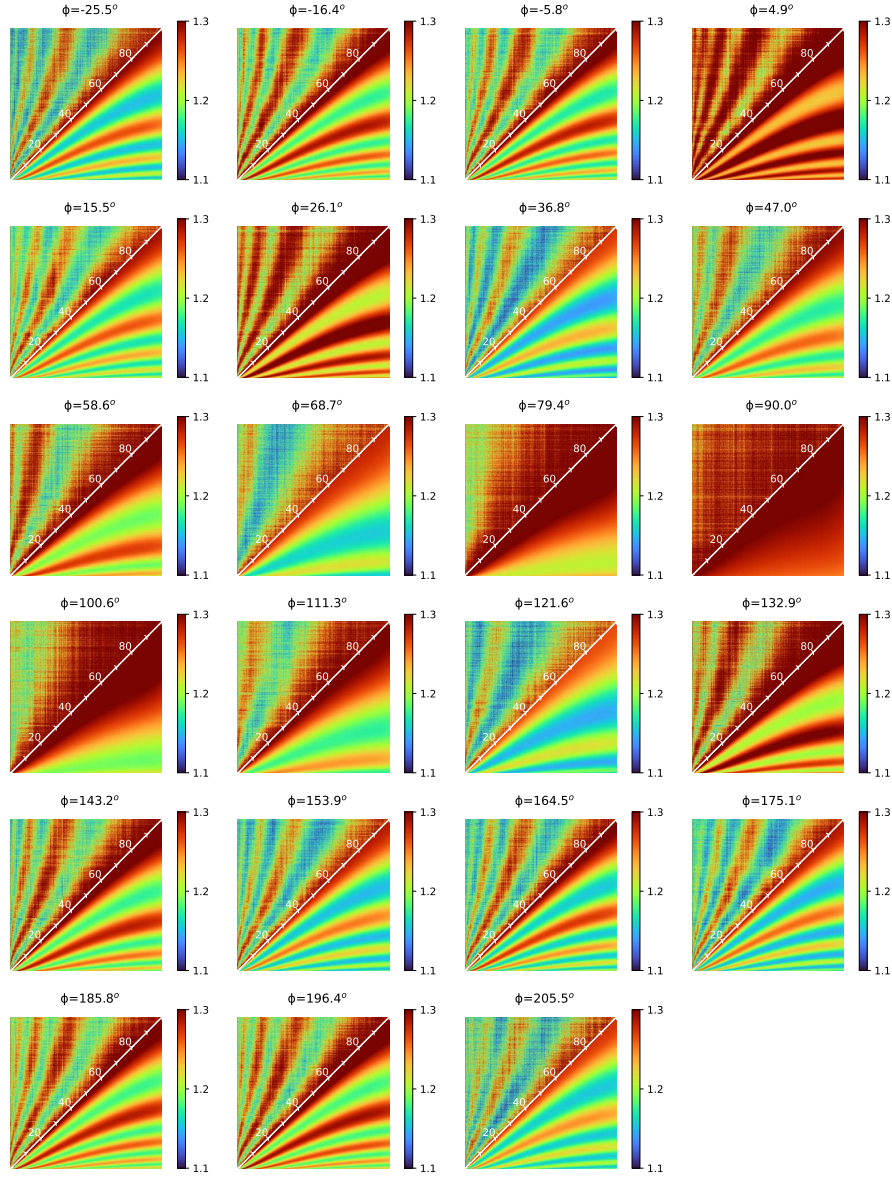


Fig. S18. $c_2(\vec{q}, t_1, t_2)$ functions for $\text{SiO}_x\text{-Cys}$ in a 100 mM NaNO_3 solution under a stress of 100 Pa, shown for all directions. Experimental results are displayed in the upper corners of each panel, with the corresponding fitting results in the lower corners.

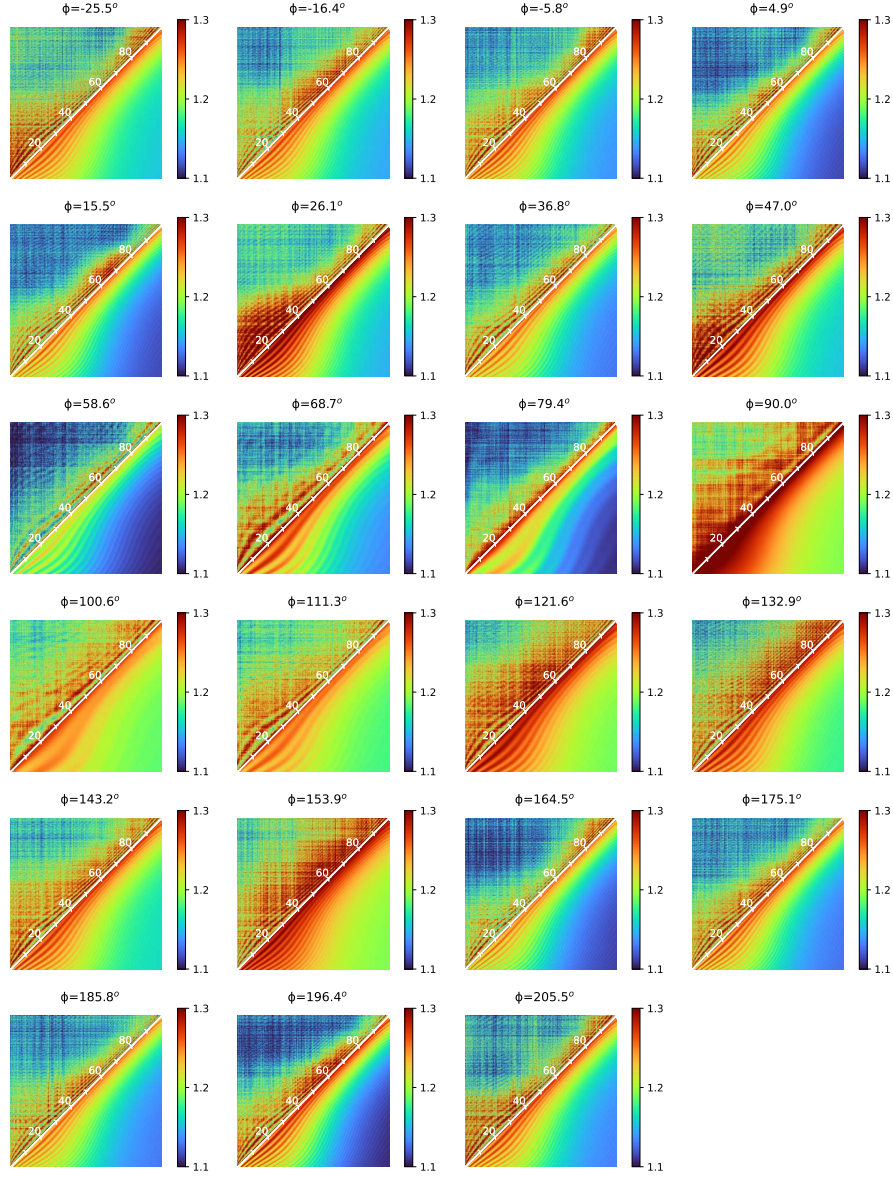


Fig. S19. $c_2(\vec{q}, t_1, t_2)$ functions for $\text{SiO}_x\text{-Cys}$ in a 100 mM NaNO_3 solution under a stress of 110 Pa, shown for all directions. Experimental results are displayed in the upper corners of each panel, with the corresponding fitting results in the lower corners.

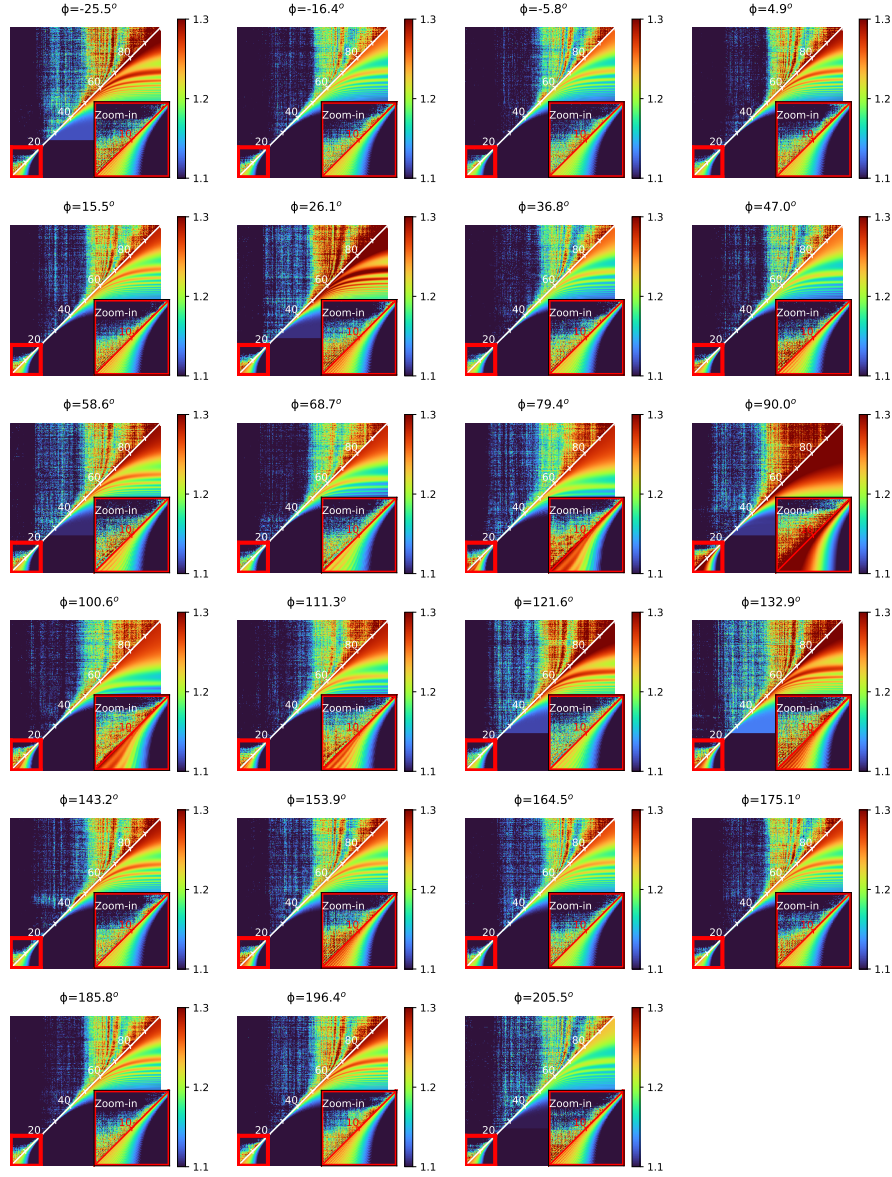


Fig. S20. $c_2(\vec{q}, t_1, t_2)$ functions for $\text{SiO}_x\text{-Cys}$ in a 100 mM NaNO_3 solution under a stress of 120 Pa, shown for all directions. Experimental results are displayed in the upper corners of each panel, with the corresponding fitting results in the lower corners.

Resolidification. In this experiment, we also investigated the yielding behavior of another salted colloidal suspension, spherical silica particles surface-functionalized by N-acetyl cysteine ($\text{SiO}_x\text{-Nac}$) (negatively charged), in 10 mM magnesium nitrate ($\text{Mg}(\text{NO}_3)_2$) and 70 mM NaNO_3 during a creep test using small angle X-ray scattering (SAXS), X-ray photon correlation spectroscopy (XPCS), and *in situ* rheology (Rheo-SAXS-XPCS). The ionic conditions were carefully controlled to match the ionic strength used for $\text{SiO}_x\text{-Cys}$. Under these conditions, $\text{Mg}(\text{NO}_3)_2$ induces both a screening and crosslinking effect, enhancing attractive interactions between the colloidal surfaces. This attraction results in complex rheological behaviors, including delayed yielding and resolidification, which are similar to those observed for $\text{SiO}_x\text{-Cys}$. These findings suggest that such rheological phenomena are common across charged colloidal systems.

At the microscopic level, resolidification occurs when the system transitions from net bond loss to bond formation. Upon cessation of flow, these bonds rapidly decrease as the force chains lock in place. However, as energy storage saturates and applied stress resumes, particle mobility is restored, leading to a reversal in bond dynamics. This process results in cycles of delayed yielding and resolidification, as demonstrated by the oscillatory shear observed in the flow rate, shown in Fig.2.

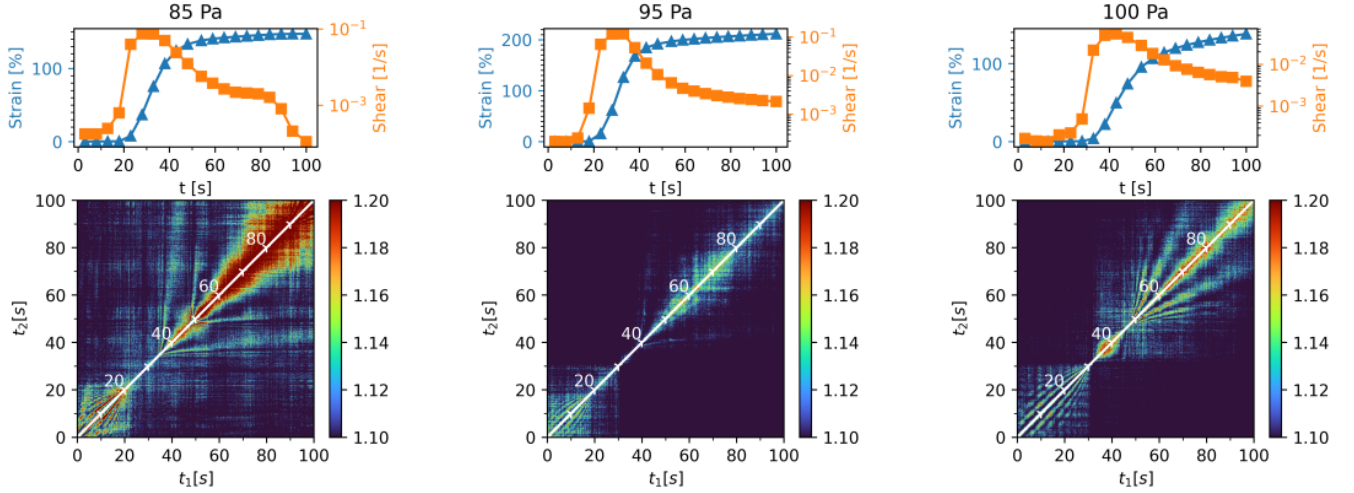


Fig. S21. Rheo-SAXS-XPCS results for $\text{SiO}_x\text{-Nac}$ during creep tests at applied stresses of 85, 95, and 100 Pa. Resolidification is observed in all trials, demonstrating that this rheological phenomenon is common in charged colloidal systems.

Delayed Yielding After Resolidification. In addition to studying neutral ($\text{SiO}_x\text{-Cys}$) and negatively charged ($\text{SiO}_x\text{-Nac}$) colloidal suspensions, we also examined a positively charged colloidal suspension, spherical silica particles surface-functionalized by L-cysteine methyl ester ($\text{SiO}_x\text{-CME}$). This suspension exhibits higher viscosity, likely due to interactions between the positively charged molecules and the negatively charged surface of SiO_x . Despite the increased viscosity, complex rheological phenomena were observed during yielding in its salted suspension (10 mM sodium nitrate (NaSO_4) and 70 mM NaNO_3).

In the creep test with Rheo-SAXS-XPCS, we observed a second instance of delayed yielding following resolidification. This twice-delayed yielding aligns with the microscale picture of junctions between shear bands. After resolidification, the rebuilt junctions are unable to sustain the applied stress, resulting in the breakdown of these junctions and the subsequent delayed yielding.

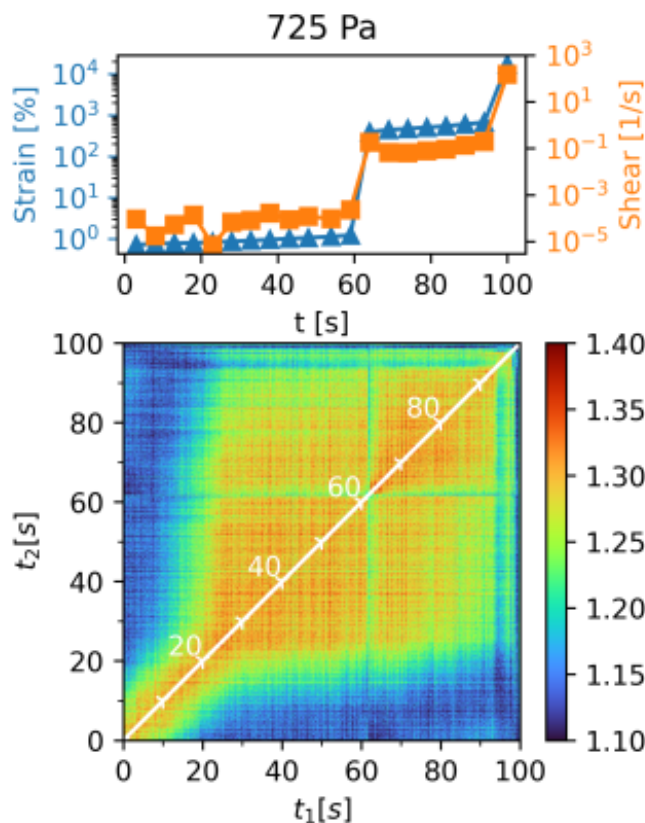


Fig. S22. Rheo-SAXS-XPCS results for $\text{SiO}_x\text{-CME}$ in a creep test at 725 Pa, showing two instances of delayed yielding following resolidification.

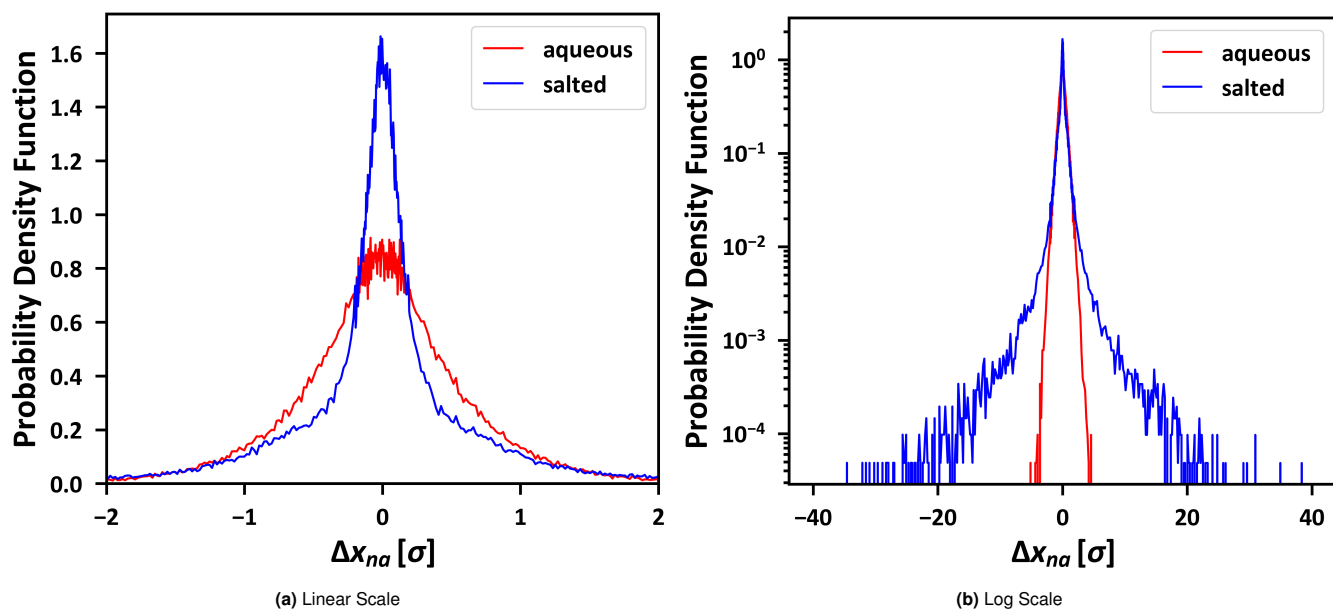


Fig. S23. Comparison of the Δx_{na} distributions in aqueous/repulsive and salted/attractive suspensions, shown in (a) linear and (b) log scale, corresponding to Fig.3D and Fig.3H, respectively. The repulsive suspension exhibits a distribution closely resembling a Gaussian profile. In contrast, the attractive suspension exhibits a distinctly non-Gaussian profile, characterized by a sharper central Gaussian peak, reflecting slow particles in a condensed phase that is denser than repulsive suspensions, and a broad tail, arising from fast particles located near the interface.

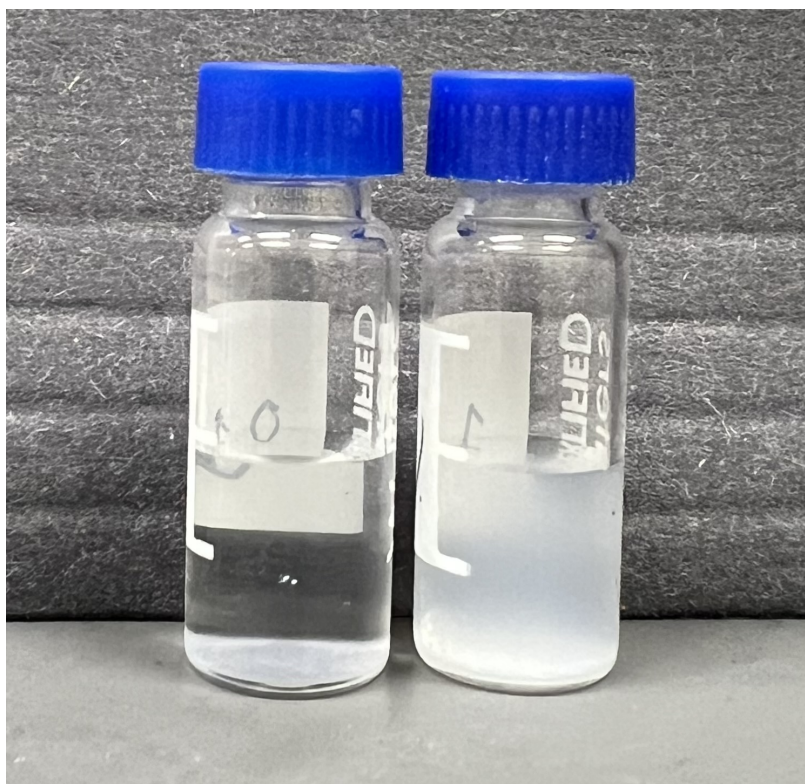


Fig. S24. Photographs of SiO_x -Cys aqueous suspensions at 0.1 vol % (left) and 1 vol % (right). The increased opacity at 1 vol % suggests the onset of coagulation near this concentration, indicating that the critical coagulation concentration (CCC) likely lies at or below 1 vol %.

145 **Supplementary Simulation Movies**

146 The following movies visualize the simulated dynamics of colloidal suspensions under shear, illustrating the impact of interparticle
147 interactions on the yielding process and the formation of shear bands. Each simulation highlights key features of the system's
148 response to applied stress, ranging from homogeneous flow in repulsive systems to complex rheological phenomena such as
149 shear banding, delayed yielding, and resolidification in attractive systems. Particles are color-coded based on their shear rate
150 to enhance visualization of flow characteristics. Detailed descriptions of each movie are provided below.

151 **Movie S1.** Shear band formation in an attractive suspension. The gray surface represents void regions initially distributed
152 uniformly. As deformation progresses, a concentrated shear band layer emerges, driven by attractive interactions that squeeze
153 voids out of the interface, forming low-density regions. These regions are mechanically weaker, leading to strain localization.

154 **Movie S2.** Yielding process of a repulsive suspension with repulsive interactions under a stress of 0.1. Particles are
155 color-coded by their shear rate. Due to the repulsive potential, the shear rate remains homogeneously distributed, exhibiting a
156 linear profile across the gap, characteristic of laminar flow.

157 **Movie S3.** Yielding process of an attractive suspension with attractive interactions under a stress of 1.3. Particles are
158 color-coded by their shear rate. The attractive potential induces shear banding, where distinct bands exhibit different flow
159 velocities. During yielding, the junction between the shear bands becomes critical, as the breaking and reforming of this
160 junction governs delayed yielding and resolidification.

References

1. Y Liu, WR Chen, SH Chen, Cluster formation in two-yukawa fluids. The J. chemical physics **122** (2005).
2. AP Thompson, et al., LAMMPS - a flexible simulation tool for particle-based materials modeling at the atomic, meso, and continuum scales. Comp. Phys. Comm. **271**, 108171 (2022).
3. A Kumar, JJ Higdon, Origins of the anomalous stress behavior in charged colloidal suspensions under shear. Phys. Rev. E - Stat. Nonlinear, Soft Matter Phys. **82**, 051401 (2010).
4. A Kumar, Ph.D. thesis (University of Illinois at Urbana-Champaign) (2010).
5. R Ball, JR Melrose, A simulation technique for many spheres in quasi-static motion under frame-invariant pair drag and brownian forces. Phys. A: Stat. Mech. its Appl. **247**, 444–472 (1997).
6. DS Bolintineanu, et al., Particle dynamics modeling methods for colloid suspensions. Comput. Part. Mech. **1**, 321–356 (2014).
7. L Banetta, F Leone, C Anzivino, MS Murillo, A Zaccone, Microscopic theory for the pair correlation function of liquidlike colloidal suspensions under shear flow. Phys. Rev. E **106**, 044610 (2022).